Publications List:

Journal Publications:

1- U.K.A. Klein, J. Mastromarino, **Z. Yamani** and A. Suwaiyan, "Fluorescence demodulation spectroscopy. A new method of determining fluorescence decays using two beating ring dye lasers". Chem. Physics Letters. **217**(1-2), 80-85 (1994)

2- W.H. Thompson, **Z. Yamani**, L.H. Abu-Hassan, J. Green, M. Nayfeh and M-A Hasan, "Room temperature oxidation enhancement of porous Si(001) using ultraviolet-ozone exposure". J. of Appl. Phys. **80**(9), 5415 (1996)

3- N. Rigakis, **Z. Yamani**, L.H. Abu-Hassan, J. Hilliard and M.H. Nayfeh, "Time-resolved measurements of the photoluminescence of Cu-quenched porous silicon". Appl. Phys. Lett. **69**(15), 2216-2218 (1996)

4- **Zain Yamani**, W.Howard Thompson, Laila Abu Hassan and Munir Nayfeh, "Ideal anodization of silicon". Appl. Phys. Lett. **70**(25), 3404 (1997)

5- M. Nayfeh, N. Rigakis and **Z. Yamani**, "Photoexcitation of Si-Si surface states in nanocrystallites". Phys. Rev. B **56**(4), 2079 (1997)

6- **Zain Yamani**, Sahel Ashhab, Ammar Nayfeh, W. Howard Thompson and Munir Nayfeh, "Red to Green Rainbow Photoluminescence from unoxidized silicon nanocrystallites". J. of Appl. Phys. **83**(7), 3929 (1998)

7- **Z. Yamani**, N. Rigakis, and M.H. Nayfeh, "Excitation of size selected nanocrystallites in porous silicon". Appl. Phys. Lett. **72**(20), 2556 (1998)

8- W.H. Thompson, **Z. Yamani**, L. Abu-Hassan, O. Gurdal and M. Nayfeh, "The effect of ultrathin oxides on luminescent silicon nanocrystallites". Appl. Phys. Lett. **73**(6), 841 (1998)

9- L. H. Abu-hassan, A. J. Abu El-Haija, S. Mahmood, **Z. Yamani**, M.H. Nayfeh, "Structural characterization of porous silicon as a function of depth". Dirasat: Natural and Engineering Sciences, **25**(3), 427-433 (1998)

10- **Z.H. Yamani**, A. Alaqal, J. Therrien, O. Nayfeh and M. Nayfeh, "Revival of interband crystalline reflectance from nanocrystallites in porous silicon by immersion plating". Appl. Phys. Lett. **74**(23), 3483-3485 (1999)

11- **Z.H. Yamani**, O. Gurdal, A. Alaql and Munir Nayfeh, "Correlation of diffuse scattering with nanocrystallite size in porous silicon using transmission microscopy". J. Appl. Phys. **85**, 8050 (1999)

12- M. Nayfeh, O. Akcakir, J. Therrien, **Z.H, Yamani**, N. Barry, W. Yu, and E. Gratton, "Highly nonlinear photoluminescence threshold in porous silicon". Appl. Phys. Lett. **75**(26), 4112 (1999)

13- M.A Dastageer, **Z.H. Yamani** and F.F. Al-Adel, "The Collisional Cooling Effect of Different Carrier Gases on a Selected ν_2" Hot Band of Sulphur Dioxide", Asian Journal of Spectroscopy **4**(4), 173 (2000)

14- M. A. Gondal, A. Dastgeer, **Zain. H. Yamani**, M. A. Arfaj and M.A. Ali, "Laser-induced fluorescence monitoring of higher alkanes production from pure methane using non-oxidative processes", Talanta, **59**(2), 295-302 (2003)

15- M.A. Gondal, **Z.H. Yamani** A. Dastageer, M.A. Ali and A. Arfaj, "Photo-conversion of Methane into Higher Hydrocarbons Using 355 nm Laser Radiation", Spectroscopy Letters, **36**(4), 313-326 (2003)

16- M.A. Gondal, A. Dastageer, **Z.H. Yamani**, A. Arfaj "Investigation of Stimulated Raman Scattering of ν_1 and ν_2 Modes in CH_4", Chemical Physics Letters, **377**(1-2), 249-255 (2003)

17- Gondal, M. A., A. Hameed, **Z.H. Yamani**, and A. Al-Suwaiyan, Production of Hydrogen and Oxygen by Water Splitting Using Laser Induced Photo-Catalysis over Fe_2O_3, Applied Catalysis **268**(1-2), 159-167 (2004).

18- M.A. Gondal, A. Hameed, **Z.H. Yamani** and A. Al-Suwaiyyan, Laser Induced Photo-catalytic Oxidation/Splitting of Water over α-Fe_2O_3, WO_3, TiO_2 and NiO Catalysts: Activity Comparison, Chemical Physics Letters , **385**(1-2), 111-115 (2004)

19- M. A. Gondal, A Hameed, **Z. H. Yamani** and A. Arfaj, Photocatalytic Transformation of Methane into Methanol under UV Laser Irradiation over WO_3, TiO_2 and NiO Catalysts, Chemical Physics Letters **392**(4-6), 372-377(2004).

20- M. A.Gondal, A. Hameed , **Z. H. Yamani**, Hydrogen Generation by Laser Transformation of Methanol using n-type WO_3 Semiconductor Catalyst, J. Molecular Catalysis A, **222**(1-2), 259-264 (2004).

21- A. Hameed , M. A. Gondal, **Z.H. Yamani**, Effect of Transition Metal Doping on Photocatalytic Activity of WO_3 for water splitting under Laser Illumination: Role of 3d-orbitals. Catalysis Communication. **5**(11), 715-719 (2004).

22- A. Smith, **Z. H. Yamani**, J. Turner, N. Roberts, S. R. Habbal, S. Granick, and M. H. Nayfeh , Observation of Strong Direct-Like Oscillator Strength in the Photoluminescence of Si_{29} Nanoparticles Physical Review B **72**(20), 205307/1-205307/5 (2005).

23- A. Hameed, Gondal, M. A., **Z. H. Yamani** and A.H. Yahya, Significance of pH Measurements in Photocatalytic Splitting of Water using 355 nm UV Laser, J. Molecular Catalysis A. **227**(1-2), 241-246 (2005).

24- **Zain H. Yamani**, "Clean Production of Hydrogen Via Laser-Induced Methane Conversion", Energy Sources, **27**(8), 661-668 (2005).

25- M. A.Gondal, A. Hameed , **Z. H. Yamani**, Laser Induced Photocatalytic Splitting of Water over WO_3 Catalyst, Energy Sources, **27**(12), 1151-1165, (2005).

26- Gondal, M. A., T. Hussain, **Z.H. Yamani** and Z. Ahmed. Determination of Toxic Metals in Petroleum, Cultivated Land and Ore Samples Using Laser-Induced Breakdown Spectroscopy, Bulletin Of Environmental Contamination & Toxicology, **78**(3-4), 270-274 (2007).

27- Gondal, M. A, T. Hussain, **Z.H. Yamani** and A.H. Bakry, Study of hazardous metals in iron slag waste using laser induced breakdown spectroscopy, J. Environmental Science and Health, part-A, **42**(6), 767-775 (2007).

28- Gondal, M. A, T. Hussain, **Z.H. Yamani** and M. A. Baig, The Role of Various Binding Materials for Trace Elemental Analysis of Powder Samples Using Laser Induced Breakdown Spectroscopy, Talanta , **72**(2), 642-649 (2007).

29- Gondal, M. A, **Z. H. Yamani**, Highly Sensitive Electronically Modulated Photoacoustic Spectrometer for Ozone Detection, Applied Optics **46**(29), 7083-7090 (2008).

30- Gondal, M. A., Hussain, T., **Z. H. Yamani**, Optimization of the LIBS Parameters for Detection of Trace Metals in Petroleum Products. Energy Sources, Part A: Recovery, Utilization, and Environmental Effects, **30**(5), 441-451 (2008).

31- Gondal, M. A, A. Dastageer, **Z. H. Yamani**. Laser Induced Photoacoustic Detection of Ozone at 266 nm Using Resonant Cells of Different Configuration, J. Environment Science and Health Part A Vol.**44**(13), 1457-64 (2009).

32- Gondal, M. A. T. Hussain, **Z. H. Yamani** and M.A. Baig. On line Monitoring of Remediation Process of Chromium Contaminated Soil using Laser Induced Breakdown Spectroscopy, J. Hazardous Materials, **163**, 1265-1271 (2009).

33- Gondal, M. A, M.N. Sayeed, **Z.H. Yamani** and A. Arfaj. Efficient Removal of Phenol from Water Using Fe2O3 Semiconductor Catalyst Under UV Laser Irradiation, J. Environment Science and Health Part A Vol. A, **44**(5), 515-521 (2009).

34- Gondal, M. A , Z. Ahmad, A Nasr, **Z.H. Yamani**, Determination of Trace Elements in Volcanic Rock Samples Collected From Cenozoic Lava Eruption Sites Using LIBS, J. Environment Science and Health Part A Vol. A, **44**(5), 528-535 (2009).

35- Gondal, M. A., Q.A. Drmosh, **Z.H. Yamani** and M. Rashid, Synthesis of Nanostructure ZnO and ZnO_2 by Laser Ablation Process Using Third Harmonic of Nd: YAG Laser, International Journal of Nano Particles, **2**, 119-128 (2009).

36- Gondal, M. A. T. Hussain, **Z. H. Yamani** and O.S.B. Al-Amoudi, Spectral Determination of Chloride Content in Different Types of Cement Using LIBS, Spectroscopy Letters, **42**, 171-177 (2009).

37- M. Qamar, M.A. Gondal, K. Hayat , **Z.H. Yamani**, K. Al-Hooshani, Laser-induced removal of a dye C.I. Acid Red 87 using n-type WO_3 semiconductor catalyst, J. Hazardous Materials , **170**, 584-589 (2009).

38- M. Qamar, M.A. Gondal and **Z.H. Yamani**, Synthesis of highly active nanocrystalline WO_3 and its application in laser-induced removal of a dye from water, Catalysis Communications, **10** (15), 1980-1984 (2009).

39- M.A. Gondal, Q.A. Drmosh, **Z.H. Yamani** and T.A. Saleh , Synthesis of ZnO_2 nanoparticles by laser ablation in liquid and their annealing transformation into ZnO nanoparticles, Applied Surface Science, **256** (1), 298-304 (2009).

40- M. Qamar, M.A. Gondal, **Z.H. Yamani** Removal of Rhodamine 6G induced by laser and catalyzed by Pt/WO_3 nanocomposite, Catalysis Comm.Volume **11**, 768-772 (2010).

41- Q. A. Drmosh, M.A. Gondal, **Z.H. Yamani** and T.A. Saleh, Spectroscopic Characterization Approach to Study Surfactants Effect On ZnO_2 Nanoparticles Synthesis by Laser Ablation Process, Applied Surface Science, **256**, pp. 4661-4666 (2010).

42- Nacir Tit , **Z.H.Yamani**, J. Graham, A. Ayesh, Origins of visible-light emissions in hydrogen-coated silicon nanocrystals: Role of passivating coating, Journal of Luminescence **130**, 2226–2237 (2010).

43- Nacir Tit, **Zain H. Yamani**, John Graham, Ahmad Ayesh, Effects of the passivating coating on the properties of silicon nanocrystals, Materials Chemistry and Physics, **124**, 927–935 (2010).

44- Ahmed A.I., Khalil , Fathy, Salman , **Zain, Yamani**, UV laser-induced transport propertieschange in silver metaphosphateglass, Optics Communications, **283**(24), 5173-5182 (2010).

45- M.A. Gondal, X.F. Chang, **Z.H. Yamani**, UV-light induced photocatalytic decolorization of Rhodamine 6G molecules over BiOCl from aqueous solution, Chemical Engineering Journal, **165**, 250–257 (2010).

46- Gondal, M.A., Dastageer, M.A., Khalil, A., Hayat, K., and **Yamani, Z.H.**, Nanostructured ZnO synthesis and its application for effective disinfection of Escherichia coli micro organism in water, Journal of Nanoparticle Research, **13**(8), 3423-3430 (2011).

47- Hayat, K., Gondal, M.A., Khaled, M.M., **Yamani, Z.H.**, and Ahmed, S**.,** "Laser induced photocatalytic degradation of hazardous dye (Safranin-O) using self synthesized nanocrystalline WO_3," Journal of Hazardous Materials, **186**, 1226-1233 (2011).

48- Gondal, M.A., Nasr, M.M., Ahmed, M.M., **Yamani, Z.H.**, and Al-Salhi, M.S. Detection of lead in paint samples synthesized locally using laser-induced breakdown spectroscopy**,** Journal of Environmental Science and Health Part A, **46**, 42-49 (2011).

49- Tawfik A. Saleh, M.A. Gondal. Q.A. Drmosh, **Z.H. Yamani** and A. Al-Yamani. Enhancement in Photocatalytic Activity for Acetaldehyde Removal by Embedding ZnO nano particles on Multiwall Carbon Nanotubes, Chemical Engineering Journal, **166**, 407-412 (2011).

50- Nacir Tit, **Zain H. Yamani**, Giovanni Pizzi, Michele Virgilio, Comparison of confinement characters between porous silicon and silicon nanowires. Physics Letters A, **375**, 2422–2429 (2011).

51- M. Qamar, M.A. Gondal and **Z.H. Yamani**, Laser-induced efficient reduction of Cr(VI) catalyzed by ZnO nanoparticles J. Hazardous Materials, **187**, 258–263 (2011).

52- M. Qamar, M.A. Gondal, **Z.H. Yamani**, Synthesis of nanostructured NiO and its application in laser-induced photocatalytic reduction of Cr(VI) from water, Journal of Molecular Catalysis A: Chemical, **341**, 83–88 (2011).

53- M. A. Gondal, X. Chang, **Z.H. Yamani**, G. Yang and G. JI, GaN thin films growth and their application in photocatalytic removal of sulforhodamine B from aqueous solution under UV pulsed laser irradiation, Journal of Environmental Science and Health Part A, **46**, 415–419 (2011).

54- Mohammed Ashraf Gondal, Xiaofeng Chang, Mohammad Ashraf Ali, **Zain Hassan Yamani**, Qin Zhou, Guangbin Ji, Adsorption and degradation performance of Rhodamine B over BiOBr under monochromatic 532 nm pulsed laser exposure, Applied Catalysis A, 397(1-2), 192-200 (2011).

55- M. Qamar, **Z.H. Yamani**, M.A. Gondal, K. Alhooshani, Synthesis and comparative photocatalytic activity of Pt/WO3 and Au/WO3 nanocomposites under sunlight-type excitation, Solid State Sciences, **13**, 1748-1754 (2011).

56- Xuanqi Liang, M.A. Gondal, Xiaofeng Chang, **Zain H Yamani**, Nianwu Li, Hongling Lu and Guangbin Ji. Facile preparation of magnetic separable powdered-activated-carbon/Ni adsorbent and its application in removal of perfluorooctane sulfonate (PFOS) from aqueous solution, J. Environ. Sci. Health A, **46**(13), 1482-1490 (2011).

57- Gondal, M.A., Hayat, K., Khaled, Mazen M., **Yamani, Z.H.**, Ahmed, Shakeel, Photocatalytic removal of hazardous dye from water using nanostructured WO_3, International Journal of Nanoparticles, **4**(1), 53-63 (2011).

58- M.A. Gondal, Chunli Li, Xiaofeng Chang, Lek Sikong, **Zain H. Yamani**, Qin Zhou, Fan Yang and Qin Lin, Facile preparation of magnetic C/TiO2/Ni composites and their photocatalytic performance for removal of a dye from water under UV light irradiation Journal of Environmental Science and Health, Part A 47(4), 570–576 (2012).

59- Mohammed A. Gondal, Xiaofeng Chang, Abdulaziz A. Al-Saadi, **Zain H. Yamani**, Qin Zhou, Guangbin Ji, BiOCl-assisted photodegradation of Rhodamine B under white light and monochromatic green pulsed laser irradiation, Journal of Environmental Science and Health, Part A, 47 (8), 1192-1200 (2012).

60- Gondal, M. A., Ali, M., Chang, X. F., Shen, K., Xu, Q. Y., **Yamani, Z. H.**, Pulsed laser-induced photocatalytic reduction of greenhouse gas CO_2 into methanol: A value-added hydrocarbon product over SiC, Journal of Environmental Science and Health, Part A 47(11), 1571-1576 (2012).

61- M. Qamar, **Z.H. Yamani**, Bismuth oxychloride-mediated and laser-induced efficient reduction of Cr(VI) in aqueous suspensions, Applied Catalysis A, 439-440, pp. 187-191 (2012).

62- Nacir Tit, **Zain H. Yamani**, Giovanni Pizzi, and Michele Virgilio, Origins of visible-light emissions in porous silicon, Phys. Status Solidi C 9(6), 1458-1461 (2012).

63- Elhalawany, Noha; Maximenko, Yulia; **Yamani, Zain H.**; Yau, Siu-Tung; Nayfeh, Munir. Soluble silicon nanoparticles-polyaniline capsules for bio-sensing and imaging. J. Mater. Res., Vol. 28, No. 2, Jan. 28, 2013.

64- Gondal., M.A., Chang, X, Sha, W.E.I., **Yamani, Z.H.**, and Zhou, Q., "Enhanced photoactivity on Ag/Ag3PO4 composites by plasmonic effect," Journal of Colloid and Interface Science, Vol. 392, Jan. 2013, pp. 325–330.

65- Shen, K., Gondal, M.A., Li, Z., Li, L., Xu, Q., **Yamani, Z.H**., "450 nm visible light-induced photosensitized degradation of Rhodamine B molecules over BiOBr in aqueous solution", *Reaction Kinetic Mech Cat*, Vol. 109, 2013, pp. 247–258.

66- Muraza, O., Bakare, I. A., Tago, T., Konno, H., Adedigba, A., Al-Amer, A.M., **Yamani, Z.H.**, Masuda, "Controlled and rapid growth of MTT zeolite crystals with low-aspect-ratio in a microwave reactor", T.*Chemical Engineering Journal*, Volume 226, 2013, Pages 367-376.

67- Qasem A. Drmosh, Saleem G. Rao, **Zain H. Yamani**, Mohammed A. Gondal. Crystalline nanostructured Cu doped ZnO thin films grown at room temperature by pulsed laser deposition technique and their characterization. Applied Surface Science, 270 (2013)104-108.

68- Hegde, V.I., Shirodkar, S.N., Tit, N., Waghmare, U.V., and **Yamani, Z.H.**, "First principles analysis of graphene and its ability to maintain long-ranged interaction with H_2S", *Surface Science* Vol. 621, March 2014, pp. 168–174.

69- M. Qamar, B. Merzougui, D. Anjum, A.S. Hakeem, **Z.H. Yamani**, D. Bahnemann. Synthesis and photocatalytic activity of mesoporous nanocrystalline Fe-doped titanium dioxide. Catalysis Today 230, July 2014, 158–165.

70- Zhijian Li, Mohammed Ashraf Gondal, **Zain Hassan Yamani**, Preparation of magnetic separable $CoFe_2O_4$/PAC composite and the adsorption of bisphenol A from aqueous solution, Journal of Saudi Chemical Society, 18(3), July 2014, 208–213.

71- Mohammed Sanhoob, Oki Muraza, **Zain H. Yamani**, Eid M. Al-Mutairi, Teruoki Tago, Belabbes Merzougui, Takao Masuda, Synthesis of ZSM-12 (MTW) with different Al-source: Towards understanding the effects of crystallization parameters, Microporous and Mesoporous Materials. Volume 194, August 2014, 31–37.

72- Oki Muraza, Idris A. Bakare, Teruoki Tago,Hiroki Konno, Taichi Taniguchi, Adnan M. Al-Amer, **Zain H. Yamani**, Yuta Nakasaka, Takao Masuda, Selective catalytic cracking of n-hexane to propylene over hierarchical MTT zeolite, Fuel, 135, 1 November 2014, 105–111.

73- Shan Shi, M.A. Gondal, S.G. Rashid, Qi Qi, A.A. Al-Saadi, **Z.H. Yamani**, Yihe Sui, Qingyu Xu, Kai Shen, Synthesis of g-C3N4/BiOClxBr1− x hybrid photocatalysts and the photoactivity enhancement driven by visible light , Colloids and Surfaces A: Physicochemical and Engineering Aspects, 461, 5 November 2014, 202– 211.

74- Hafeezullah, **Zain H. Yamani**, Javed Iqbal, Ahsanulhaq Qurashi, Abbas Hakeem, Rapid sonochemical synthesis of In_2O_3 nanoparticles their doping optical, electrical and hydrogen gas sensing properties. Journal of Alloys and Compounds, 616, 15 December 2014, 76–80.

75- M. A. Gondal, A. A. Adesida, S. G. Rashid, Shan Shi, Rashid Khan, **Z. H. Yamani**, Kai Shen, Qingyu Xu, Zaki S. Seddigi & Xiaofeng Chang. Preparation of WO3/g-C3N4 composites and their enhanced photodegradation of contaminants in aqueous solution under visible light irradiation. Reaction Kinetics, Mechanisms and Catalysis. 113(1) 2014. DOI 10.1007/s11144-014-0787-y

76- Gondal, Mohammed; Qahtan, Talal; Dastageer, Mohamed; **Yamani, Zain**; Anjum, Dalver, A Rapid and Cost-Effective Laser based Synthesis of High Purity Cadmium Selinide Quantum Dots. Journal of Nanoscience and Nanotechnology, 15, 1-6, 2015

77- O. Muraza, A. Adedigba, T. Tago, A.B.D. Nandiyanto, H. Konno, **Z.H. Yamani**, T. Masuda, Microwave-assisted hydrothermal synthesis of submicron

ZSM-22 zeolites and their applications in light olefin production, *Microporous and Mesoporous Materials*, Accepted December 19, 2014.

Conferences Proceedings:

1- W.H. Thompson, **Z. Yamani**, H.M. Nayfeh, M-A Hasan, J. Green and M.H. Nayfeh, "Growth of Germanium on porous Si (001)". MRS Symp. Proc. **452**, 255 (1997).

2- M. Nayfeh, N. Rigakis and **Z. Yamani**, "Photoexcitation of Si-Si radiative surface states in nanocrystallites". MRS Symp. Proc. **486**, 243 (1998).

3- Munir H. Nayfeh , **Zain Yamani**, Osman Gurdal and A.A. Alaql. Nanostructure of Porous Silicon Using Transmission Microscopy: Observation of Restructured Nanoclusters. MRS Symp. Proc. **536**, 191-196. (1999).

4- **Z. Yamani**, W. Howard Thompson, N. Rigakis, and M. H. Nayfeh. Excitation and Luminescence of Size Selected Nanoclusters in Porous Silicon. MRS Symp. Proc. (1999).

5- M.A. Gondal, A. Dastageer, **Z.H. Yamani**, M.A. Ali, A. Arfaj, "355-nm Photodissociation of CH_4 and Production of Hydrogen", CLEO Technical Digest Series **56**, 462 (2001).

6- Gondal, M. A., A. Dastgeer, **Zain. H. Yamani** and A. Arfaj, Photoconversion of Methane into higher hydrocarbons and hydrogen using laser, Third Jordanian International Conference of Chemistry, Yarmouk. 22–26 April, Jordan. (2002).

7- M.A. Gondal, A. Dastageer, **Z.H. Yamani**, M.A. Ali, A. Arfaj, "Characterization of stimulated Raman scattering of methane", poster presentation at the International Conference on Laser Probing (LAP-2002, Belgium).

8- M.A. Gondal, **Z.H. Yamani**, A. Dastageer, M.A. Ali, A. Arfaj, "Photoinduced coupling of methane using a pulsed UV laser". 5th International Conference and Exhibition on Chemistry in Industry, Bahrain 14-16 Oct. 2002.

9- Zaki S. Seddigi, Basel Abu Sharkh, Mustafa Achoui, Mohammed Fettouhi, Mustafa Hariri, Mazen Khaled, Ali T. Lounici, Mohammed Makkawi, Mohammed Samman, **Zain H. Yamani,** "Obstacles and Possible Solutions", Discussion Forum on: Research at KFUPM : Challenges and Opportunities, DAD-KFUPM, May 2003.

10- M.A. Gondal, A. Hameed, **Zain H. Yamani** and A. Suwaiyan, "Photocatalytic Splitting of Water into Hydrogen and Oxygen Using Laser", Petrotech-Bahrain (2003).

11- **Z.H. Yamani**, M.A. Gondal, A. Hameed, A. Dastgeer, and A. Arfaj, "Hydrogen Production by Photo-dissociation of Methane with UV Laser", Petrotech-Bahrain (2003).

12- **Z. Yamani,** M.A. Gondal, E. Hegazi, H.M. Masoudi, Industry Oriented Laser Research at KFUPM, 1st Saudi Physical Society Meeting, Dec. 2003. (Abha, Saudi-Arabia).

13- **Zain H. Yamani**, "Compositional Study of Different Currency Coins Using Non-Destructive Laser Induced Breakdown Spectroscopy", 2nd SASC, Jeddah (2004).

14- M. A. Gondal, A. Hameed, **Z. H. Yamani**, A. Suwaiyan and A. Arfaj **,** "Laser Photocatalytic Conversion of Methanol into Hydrogen", 2nd SASC, Jeddah (2004).

15- M. A. Gondal, A. Hameed, **Z. H. Yamani**, A. Suwaiyan and A. Arfaj, Activity of WO_3, NiO and Fe_2O_3 Catalysts for Hydrogen Production under UV Laser Irradiation, Chemistry in industry (CHEMINDIX 2004), Manama, Bahrain (September 27-29, 2004).

16- M. A. Gondal**,** A. Hameed, **Z. H. Yamani**, and A. Arfaj (2004), Comparative study of laser photo-catalytic conversion of methane into methanol over different catalysts, International Conference on Chemistry and Industry, Riyadh (27.11-1.12, 2004).

17- M. A. Gondal, A. Hameed, **Z. H. Yamani**, A. Suwaiyan and A. Arfaj (2004), Laser photo-catalytic conversion of methanol into hydrogen, Second Saudi Science Conference, Jeddah (Saudi Arabia).

18- A. Arfaj, M. A. Gondal, A. Hameed, and **Z.H. Yamani** (2004), Photocatalytic performance of NiO catalyst in oxidative coupling of methane into methanol under UV laser irradiation, Symposium on catalysts applications in GC countries, Sharja-UAE (13-15 December 2004).

19- د. سعد بن محمد الشهراني, **د. زين بن حسن يماني**. "تفعيل دور الإنترنت كمصدر معرفي للمخترعين الناشئة العرب" اللقاء الرابع للمخترعين السعوديين/ جامعة الملك فهد للبترول و المعادن، 12-14 صفر، 1426 هـ، 22-24 مارس، 2005 م.

20- M. A. Gondal, T. Hussain, **Z. H. Yamani** and Z. Ahmed, Environmental Applications of Laser Induced Breakdown Spectroscopy: Analysis of Oil, Soil and Ore Waste Samples. 2nd SPS Scientific Gathering, Makkah (Shawwal 20-22nd, 1426) (22-24 November 2005).

21- M. A. Gondal, A. Hameed, **Z. H. Yamani**, and Z. Seddigi (2005), Laser-induced photocatalysis and its applications in petrochemicals, fuel cells and phenol degradation , 15th Saudi-Japanies symposium on Catalysis, Dhahran (27.11-28, 11, 2005).

22- مهنّد بن جبريل أبو ديّة، عبد الله بن سعيد بازياد، عبد الله بن فهد التركي، **د. زين بن حسن يماني**، أساليب مقترحة لتطوير التعليم الجامعي في علم الفيزياء بالمملكة العربيّة السعوديّة، الملتقي العلمي الثاني للجمعية العلمية السعودية للعلوم الفيزيائية، مكة المكرمة، شوال 20-22، 1426 هـ.

23- M. A. Gondal, A. Hameed, **Z. H. Yamani**, and Z. Seddigi (2005), . Laser Induced Photocatalysis and its applications in Petrochemical, fuel cells and phenol dergradation. 15th Saudi Japanese symposium on Catalysis, Dhahran (27&28 Nov. 2005).

24- د. زين يماني، صعود فقاعة هوائية في سائل ذي لزوجة (رؤية نقدية للكتاب المقرر بخصوص قانون ستوكسز). الملتقى التربوي العلمي الثالث بالمنطقة الشرقية، 10-12 ربيع الثاني، 1427.

25- S. Abdulmajeed, T. Hussian, M. A. Gondal and **Z.H. Yamani**. Photoacoustic and laser induced breakdown spectrometry and its applications in the field of environment, 3rd Educational Scientific Gathering in the Eastern Province/ Physics Festival, Dhahran, Saudi-Arabia (May, 2006).

26- M. A. Gondal, T. Hussain, **Z. H. Yamani**. Influence Of Ambient Gas Pressure On Performance Of Laser- Induced Breakdown Spectrometry For Planetary Science Applications, 3rd Saudi Physical Society Annual Meeting , Riyadh, Saudi Arabia (December16-18, 2006).

27- M. A. Gondal, **Z. H. Yamani** and I. A. Bakhtiari. Photoacoustic detection of ozone using wave guide CO_2 laser, 14th International Conference On Photoacoustic and Photothermal Phenomena (ICPPP). Cairo, Egypt (January 6 - 9, 2007).

28- M. A. Gondal , T. Hussain, **Z. H. Yamani** and O.S.B Al-Amoudi. Determination of chloride content in different types of cements with Laser Induced Breakdown Spectroscopy, 3rd Saudi Science Conference: New Horizons in Science and Their Applications , Riyadh, Saudi-Arabia (March 10-14, 2007)

29- M. A. Gondal,T. Hussain, , **Z. H. Yamani** and A. Bakari. Determination of elemental composition in iron slag waste using laser induced breakdown spectroscopy, 3rd College of Sciences Conference, Riyadh, Saudi Arabia (March 10-14, 2007).

30- M. A. Gondal, T. Hussain, and **Z.H. Yamani**. Determination of Trace Elements in Arabian Crude Oil using Laser Radiation, International Conference on Chemistry in Industry (CHEMINDIX 2007), Manama, Bahrain (March 26-28, 2007).

31- M. A. Gondal, **Z.H. Yamani**, A. Dasatgeer and I. A. Bakhtiari. Electronically modulated photoacoustic spectrometer for detection of trace gases , International Conference on Chemistry in Industry (CHEMINDIX 2007), Manama, Bahrain (March 26-28, 2007).

32- **Zain H. Yamani**, Nanotechnology Activities at KFUPM and the Creation of the Center of Excellence in Nanotechnology (CENT). NanoKSU Conference, 16-17 Shawwal, 1428 (Riyadh).

33- M.A. Gondal, Q. Darmoosh, **Z. H. Yamani**, Synthesis of Nanostructure ZnO by Laser Ablation Process Using Third Harmonic Of Nd:YAG Laser. ICON008, Jeddah (June, 2008).

34- M. A. Gondal, Zulfiqar Ahmed, M.M. Nasr and **Z. H. Yamani**. Trace elemental analysis of lava samples using laser induced breakdown spectroscopy, 1st International Conference on Laboratory Technology (LABTECH), Manama, Bahrain (20-22 Oct, 2008).

35- M. A. Gondal, A. Dastageer and **Z.H. Yamani**. Detection Of Ozone at 266 nm Using Fourth Harmonic Of Nd:YAG Laser, 1st International Conference on Laboratory Technology (LABTECH), Manama, Bahrain (20-22 Oct., 2008).

36- M. A. Gondal, T. Hussain, and **Z.H. Yamani**. On Line Monitoring of Remediation Process of Chromium Polluted Soil Using Laser Induced Breakdown Spectroscopy (LIBS), Fourth Saudi Physical Society Annual Meeting, Riyadh, Saudi-Arabia, (November 11-12, 2008).

37- M. A Gondal,. Q.A. Drmosh, **Z.H. Yamani**. Study of Post Annealing Temperature Effect On Size of Synthesized Nano Structure Zno Using High Power Pulsed Laser Ablation Technique, The International Conference For Nanotechnology Industries The Leading Technology of 21st Century Riyadh, Saudi Arabia (5-7 April, 2009).

38- M. A. Gondal, K.Hayat, **Z.H. Yamani** and S.Ahmed. Removal Of Hazardous Organic Dye From Water Using Nanostructured Metal Oxides And Lasers, International Conference on Nano technology and Advanced material (ICNOM 2009) Bahrain (4-7 May, 2009).

39- M. A.Gondal, Q.A. Drmosh, **Z. H. Yamani**. Surfactants Effect On The Morphology Of Nanoparticles Synthesized By Novel Laser Ablation Method at 355 nm , International Conference on Nano technology and Advanced material (ICNOM 2009) Bahrain, (4-7 May, 2009).

40- M.A. Gondal, T. Hussain and **Z.H Yamani**. In-situe Monitoring of Removal of Toxic Pollutants in Soil Using Laser Induced Breakdown Spectroscopy (LIBS). KACST Proceedings (2009).

41- M.A. Gondal, M.M. Nasr, **Z.H. Yamani**, Laser Sensor for Detection of Toxic Ingredients in Cosmetic Products for Human Use, World Academy for Laser Applications (WALA), Conference Bahrain International Exhibition Centre (October 19-21, 2009).

42- M. M. Nasr, M. A. Gondal, **Z.H. Yamani**, Detection of Hazardous Contaminants in Lead based Paint using Laser Induced Breakdown Spectroscopy, Fourth Saudi Science Conference, Taybah University, Madinah, KSA (21-24 March 2010).

43- M.A. Gondal, Q.A. Drmosh , **Z.H. Yamani** and T.A. Saleh , Synthesis and Characterization of ZnO2 Nanoparticles using Optical Techniques, presented at the Fourth Saudi Science Conference, Taybah University, Madinah, KSA (21-24 March 2010).

44- M. Qamar, **Z.H. Yamani** , Synthesis and visible light photocatalytic activity of WO_3-Au nanocomposite obtained under microwave irradiation SP3–Third International Conference on Semiconductor Photochemistry, Glasgow, Scotland (12-16th April, 2010).

45- M. A. Kousa, **Z. H. Yamani**, S. M. Al-Ghamdi and S. A. Said, "Towards Leading Roles of Universities in the Community", *Higher Education International Conference*, HEIC 2010, Beirut, pp. 320-327 (May 4-6, 2010).

46- Nacir Tit, Ali Reshak, John Graham, and **Zain Yamani**, Quantum-Confinement Effects on Photoluminescence Emissions in Silicon Nanocrystals. ISCS-2010, Kagawa-Japan (May 31-June 4, 2010).

47- M. Qamar, **Z.H. Yamani**, Synthesis and photocatalytic activity of TiO2-based nanotubes obtained by alkali hydrothermal treatment. The 1st International Conference on Process Engineering and Advanced Materials (ICPEAM2010) and the 24th Symposium of Malaysian Chemical Engineers (SOMCHE2010), Kuala Lumpur, Malaysia (15-17th June, 2010).

48- M. Qamar, **Z.H. Yamani**, A comparative photocatalytic activity of Pt/WO3 and Au/WO3 nanocomposites in aqueous suspensions. 6th European Meeting on Solar Chemistry and Photocatalysis: Environmental Applications (SPEA-6), Prague, Czech Republic (13-16th June, 2010).

49- M.A. Gondal, M. Qamar, **Z.H. Yamani**, Laser induced photo-catalytic removal of chromium using ZnO semiconductor catalyst. The SEGH 2010 International conference and workshops of the Society for the Environmental Geochemistry and Health (SEGH) on Environmental quality and Health. Galway, Ireland (27th June to 2nd July, 2010).

50- Mohammed A. Gondal, Abdulkader Dastageer, Amjad B. Khalil, **Zain Hassan Yamani**, Comparative Study of Nano and Micro ZnO in Antimicrobial Activity in Water Using Laser Induced Photo-catalytic Process. The SEGH 2010 International conference and workshops of the Society for the Environmental Geochemistry and Health (SEGH) on Environmental quality and Health. Galway, Ireland (27th June to 2nd July, 2010).

51- A. A. Bagabas, M. A. Gondal, A. Khalil, M. A. Dastageer, **Z.H. Yamani**, M. M. Ashameri, Laser-Induced Photocatalytic Inactivation of Coliform Bacteria in Water Using Pd-Supported on Nano-WO3 Catalyst, 10th International Symposium on Scientific Bases for the Preparation of Heterogeneous Catalysts Louvain-la-Neuve, Belgium (July 11-15, 2010).

52- N. Tabet, M. Faiz, N. Maalej, A. Mekki A. Sunaidi and **Z. Yamani**, Nanostructured Oxides Applications in Gas Sensing and Photocatalysis. International Conference on New Materials and Active Devices (NMCA'2011). OumAl Bouaghi, Algeria (23-25 May 2011).

53- Nacir Tit, **Z.H. Yamani**, G. Pizzi and M. Virgilio, Origins of Visible-Light Emissions in Porous Silicon. 13th International Conference on Formation of Semiconductor Interfaces (2011).

54- M. Qamar, B. Merzougui, **Z.H. Yamani**, A.S. Hakeem. "Synthesis of Highly Efficient Visible-light-driven Mesoporous Fe2O3/TiO2 Nano-photocatalyst for Water Decontamination". The First International Conference on Photocatalysis and Solar Energy Conversion: Development of Materials and Nanomaterials", Daejeon, South Korea, May 29-31, 2012.

55- Adedigba Abdul-lateef, Oki Muraza, Idris Bakare, Teruoki Tago, Asep B.D. Nandiyanto, Hiroki Konno, **Zain H. Yamani**, Takao Masuda. Size Control and Rapid Synthesis of ZSM-22 Zeolite by Microwave Irradiation. International Symposium on Zeolites and Microporous Crystals (Hiroshima, Japan) July 29th - August 1st, 2012.

56- Nacir Tit, Vinay Hegde, Sharmila Shirodkar, Umesh Waghmare and **Zain Yamani**. First principles analysis of graphene and its ability to maintain long-ranged interaction with H2S, American Physical Society, APS March Meeting 2013, March 18-22, 2013

57- Yulia Maximenko, Noha Elhalawany, Kevin Mantey, **Zain Yamani**, Siu-Tung Yau, and Munir H. Nayfeh, Polyaniline–Si Nanoparticle Nanocapsules as a Dual Photovoltaic Sensitizer. Mater. Res. Soc. Symp. Proc. Vol. 1500 © 2013 Materials Research Society. DOI: 10.1557/opl.2013.182

58- Qurashi, A., Qamar, M., Hakeem, A.S., **Yamani, Z.H**., Nitrogen-doped mesoporous TiO2 nanoparticles and their photocatalytic properties, 4th International Conference on Semiconductor Photochemistry (SP4) held during 23-27th June in Prague, Czech Republic. (2013)

59- Qamar, M., Merzougui, B., **Yamani, Z.H.**, Hakeem, A.S., Anjum, D., and Bahnemann, D., Synthesis and photocatalytic activity of mesoporous nanocrystalline Fe-doped titanium dioxide, 4th International Conference on Semiconductor Photochemistry (SP4) held during 23-27th June in Prague, Czech Republic. (2013)

60- O. Muraza, T. Tago, H. Konno, A. Abdul-Lateef, **Z.H. Yamani**, T. Masuda, Catalytic Cracking of n-Hexane over Modified H-ZSM-22 Zeolite, Proceeding of 23rd Annual Saudi-Japan Symposium, Catalysts in Petroleum Refining and Petrochemicals, Dhahran, Saudi Arabia, December 2-3, 2013.

61- M. K. Hossain, Q. A. Drmosh, **Z. H. Yamani** and N. Tabet, Silver Nanoparticles on Zinc Oxide Thin Film: An insight in Fabrication and Characterization, Nanostruc2014 conf. proc., pp. 47, 2014.

62- Talal F. Qahtan, M.A. Gondal, M. A. Dastageer, **Z. H. Yamani** and D. H. Anjum Preparation of CdSe Quantum Dots using nanosecond Laser. 3rd Saudi International Meeting on Frontiers of Physics Jazan, February 18th-20th, 2014.

Book Chapters:

1. "Bismuth(V)-Containing Semiconductor Compounds and Applications in Heterogeneous Photocatalysis" **Chang, X., Gondal, M.A., Abdallah Yamani, Z.H., and Ji, G.,** in "Bismuth-Containing Compounds" Springer, Vol. 186, October 2013, pp. 343-373.

Patents:

Issued:

1- M. Nayfeh, J. Therrien, and Z. Yamani: Method for producing silicon, US 6,585,947 with a publication date of July 1, 2003.

2- M. Nayfeh, J. Therrien, and Z. Yamani: "Silicon Nanoparticle and Method for Producing the Same" 6,846,474; January 25, 2005.

3- Gondal, Muhammed A.; Pola, Josef; Yamani, Zain H.; Masoudi, Husain M.; Al-Arfaj, Abdul R. A., "Laser-based method for removal of sulfur (DMDBT) in hydrocarbon fuels". USPTO Patent Number U.S. 7871501 (Jan. 18th, 2011).

4- Gondal, M.A., Drmosh, Q., Yamani, Z.H., and Saleh, T.A.,"Method for detection of cyanide in water" Patent No. US 2011/0303555 A1, Pub. Date Dec. 15, 2011. Patent number: U.S. 8460538; Issue on June 11, 2013

Published:

5- M. A. Gondal, Z. H. Yamani, M. A. Ali, A. Al-Arfaj and M. A. Dastageer (2005) "Method for the conversion of methane into C_2 and higher hydrocarbons using UV laser". USPTO Pat. NO: US2005045467-A1 Pub. Date 3 March 2005.

6- M. A. Gondal, A. Hammed and Z.H. Yamani (2005) . "Laser Photocatalytic process for the production of hydrogen" . USPTO Pat. NO: US2005226808- A1, Pub. Date 13 Oct 2005.

7- Gondal, M.A., Drmosh, Q., Yamani, Z.H., and Saleh, T.A., "Method of Forming Zinc peroxide Nanoparticles" serial no. 12/805,218, Pub. Date Dec. 15, 2011. Patent No. US 2011/0303050 A1

8- M.H. Nayfeh and Z.H. Yamani, Magnetic and Luminescent Silicon Nanoparticles, Supermolecules and Fabriaction Methods, filed Patent on September 20, 2011 USPTO serial no. 13/237,225. Published March 29, 2012. US20120077021 A1

9- Gondal, M.A., Drmosh, Q., Yamani, Z.H., and Saleh, T.A., Method for Detection of Cyanide in Water , filed Patent on Mar 14, 2013 USPTO serial no. 13/827,999, Published Aug 8, 2013. US20130203178 A1

Filed:

10- Controlled Growth of MTT Zeolite in Microwave-Assisted Hydrothermal Synthesis. 14/151498. USPTO. January 9, 2014. Patent Pending.

11- 450 nm Visible Light-Induced Photosensitized Degradation of Rhodamine B Molecules Over BiOBr in Aqueous Solution. 14/154810. USPTO. January 14, 2014. Patent Pending.

12- Method for Producing Pt-Free Electrocatalysts for Fuel Cells and Batteries. 14/224163. USPTO. March 25, 2014.